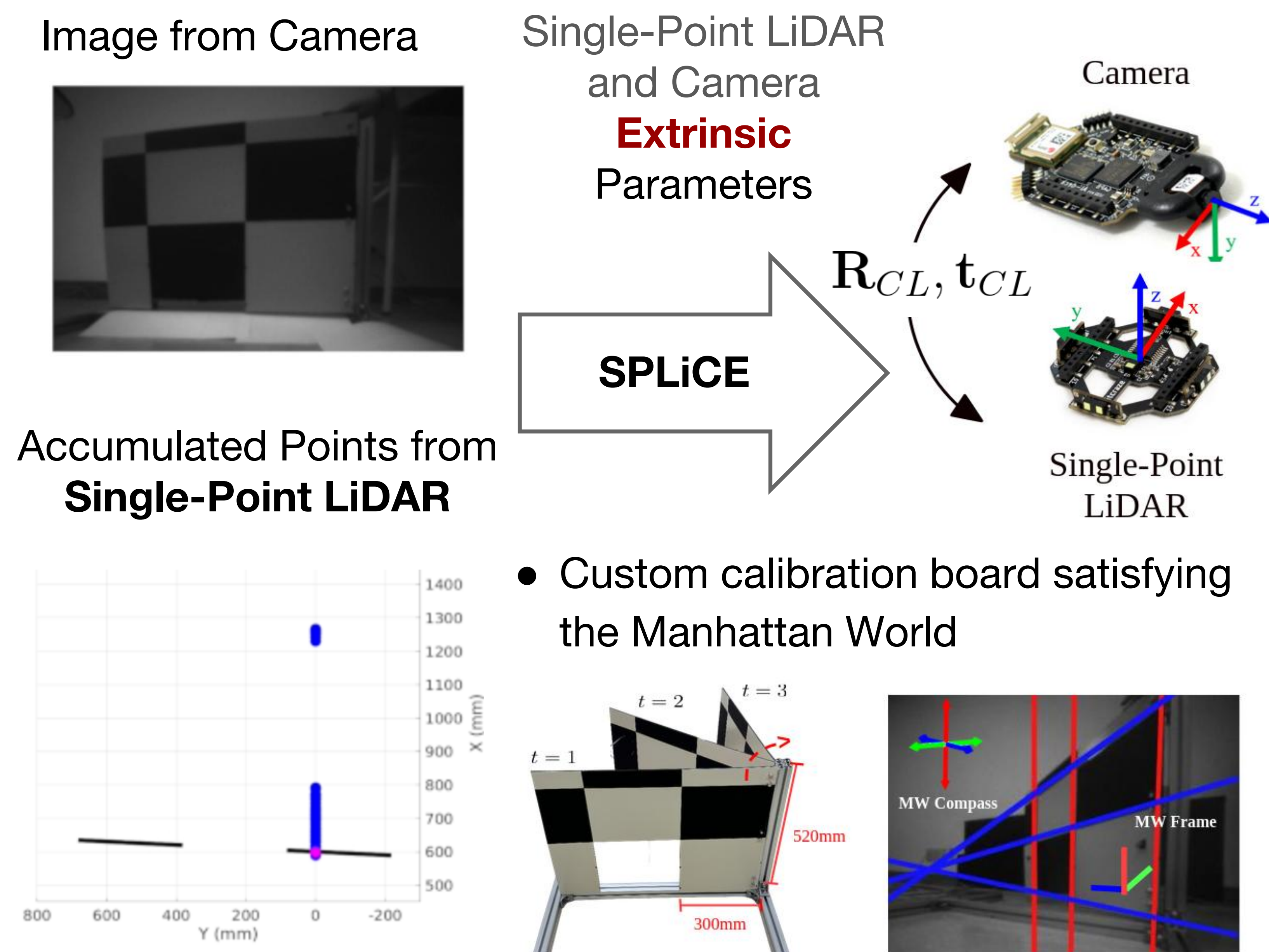
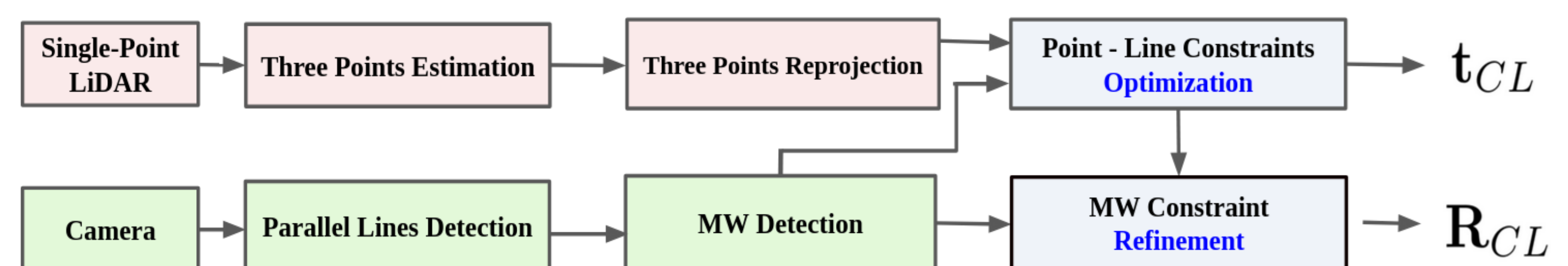


Motivation & Problem Setup

- Existing methods need **heavy, costly multi-beam LiDARs**
- Micro-robots** lack the resources to support such sensors
- Single-Point LiDAR** is lightweight but hard to calibrate



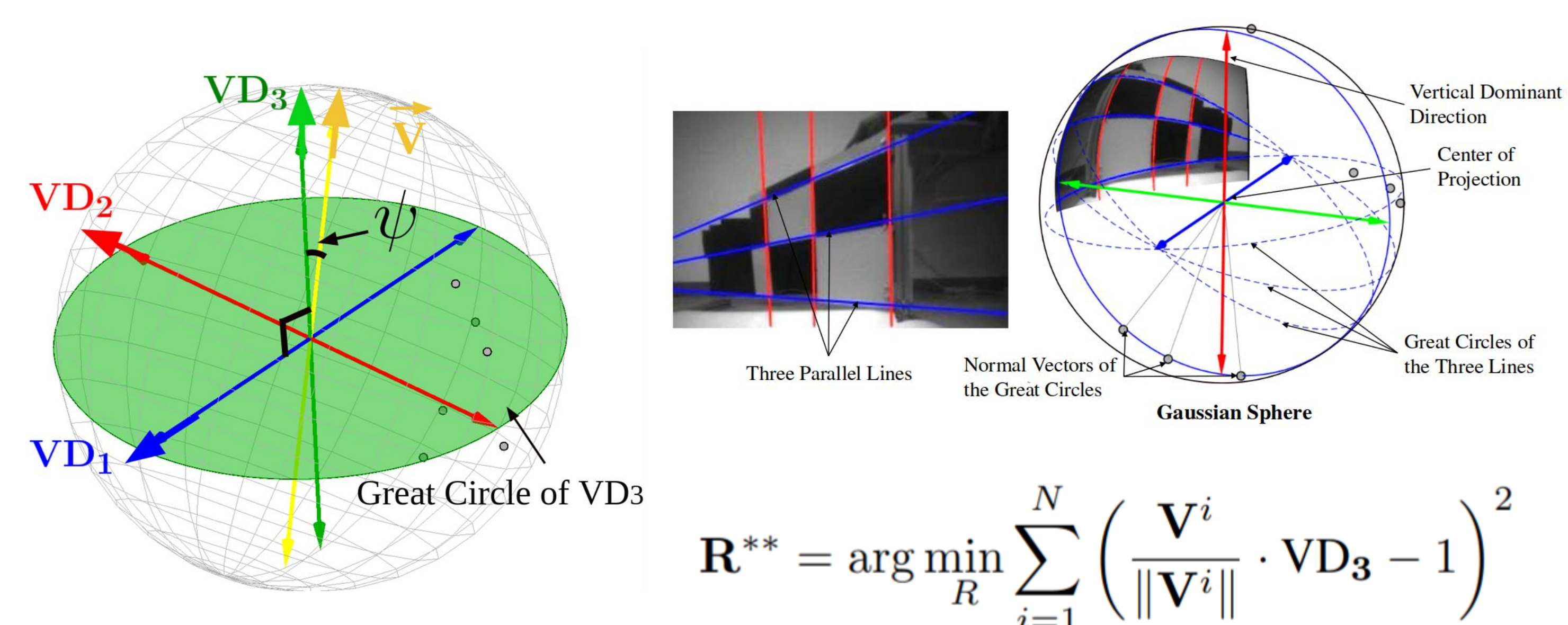
Contributions



- Custom-built** rotating calibration checkerboard
- Use of **structural regularities** with point-line features
- Evaluations & comparisons with state-of-the-art algorithms on **Crazyflie nano drone**

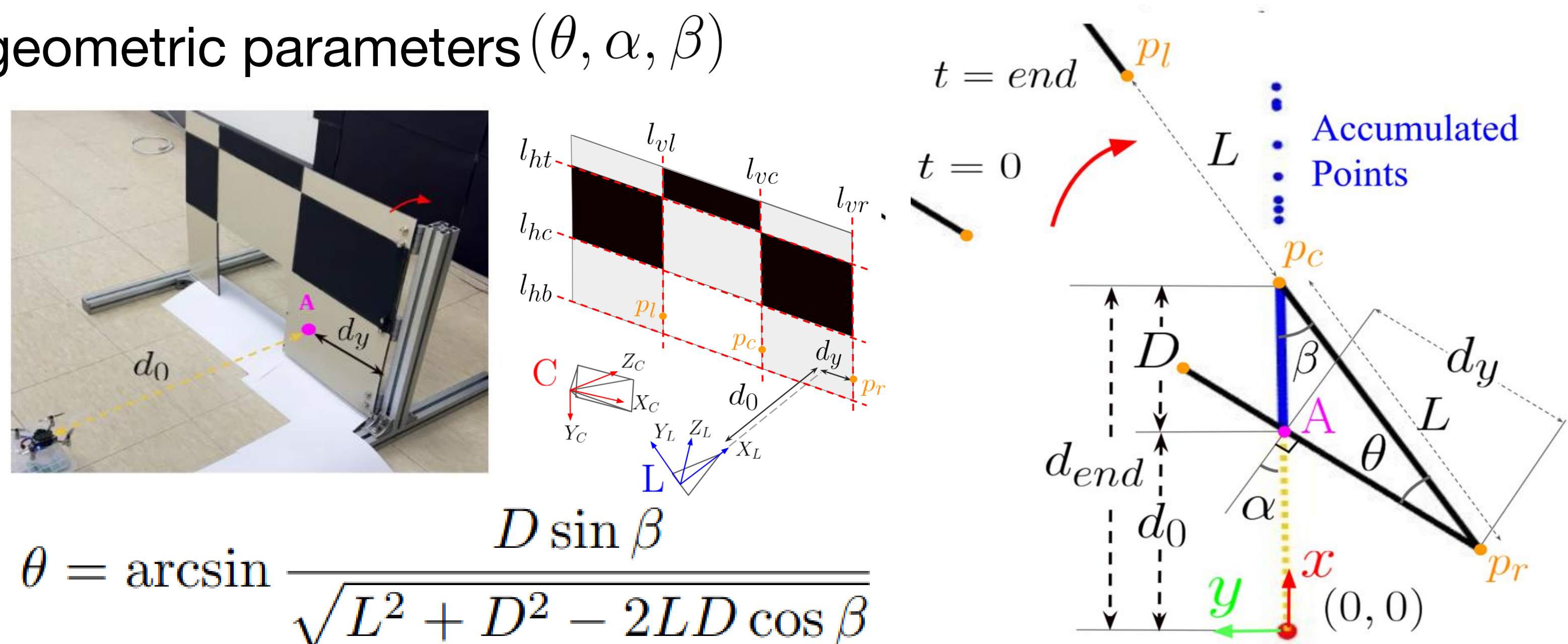
Manhattan World Refinement

- We refine the 3-DoF rotation by **aligning the board normals** with Manhattan axes



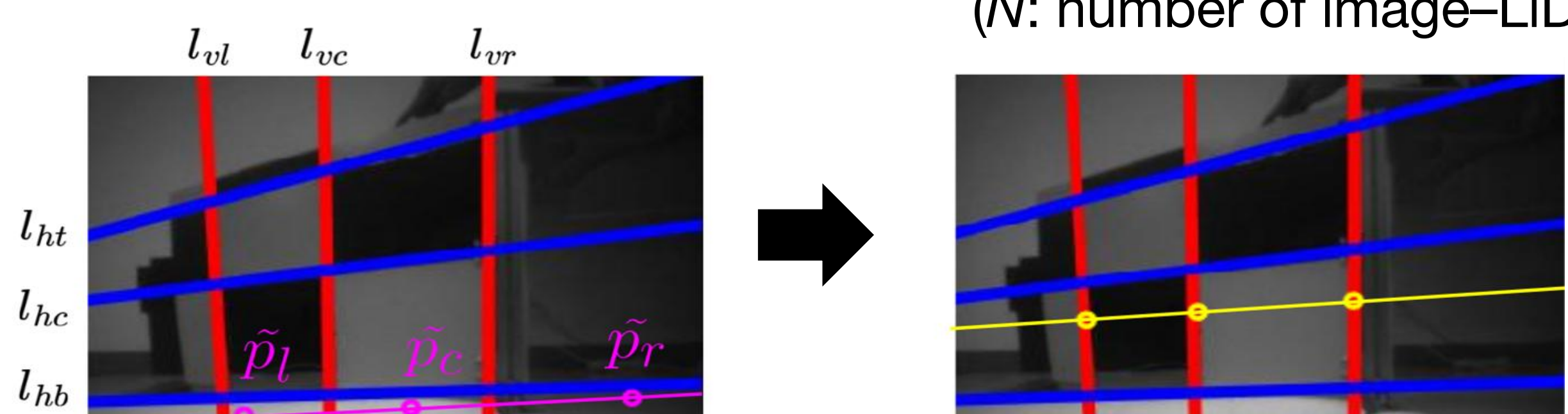
Point-Line Constraints Optimization

- We estimate **three key points** (p_r, p_c, p_l) by tracking LiDAR **distance** changes during board rotation and computing geometric parameters (θ, α, β)
- Minimizing **point-to-line distances** (d_k^i) between reprojected points ($\tilde{p}_l, \tilde{p}_c, \tilde{p}_r$) and detected vertical lines while aligning the **vanishing point** (p_j^i) with horizontal lines, optimized via the **LM** method.



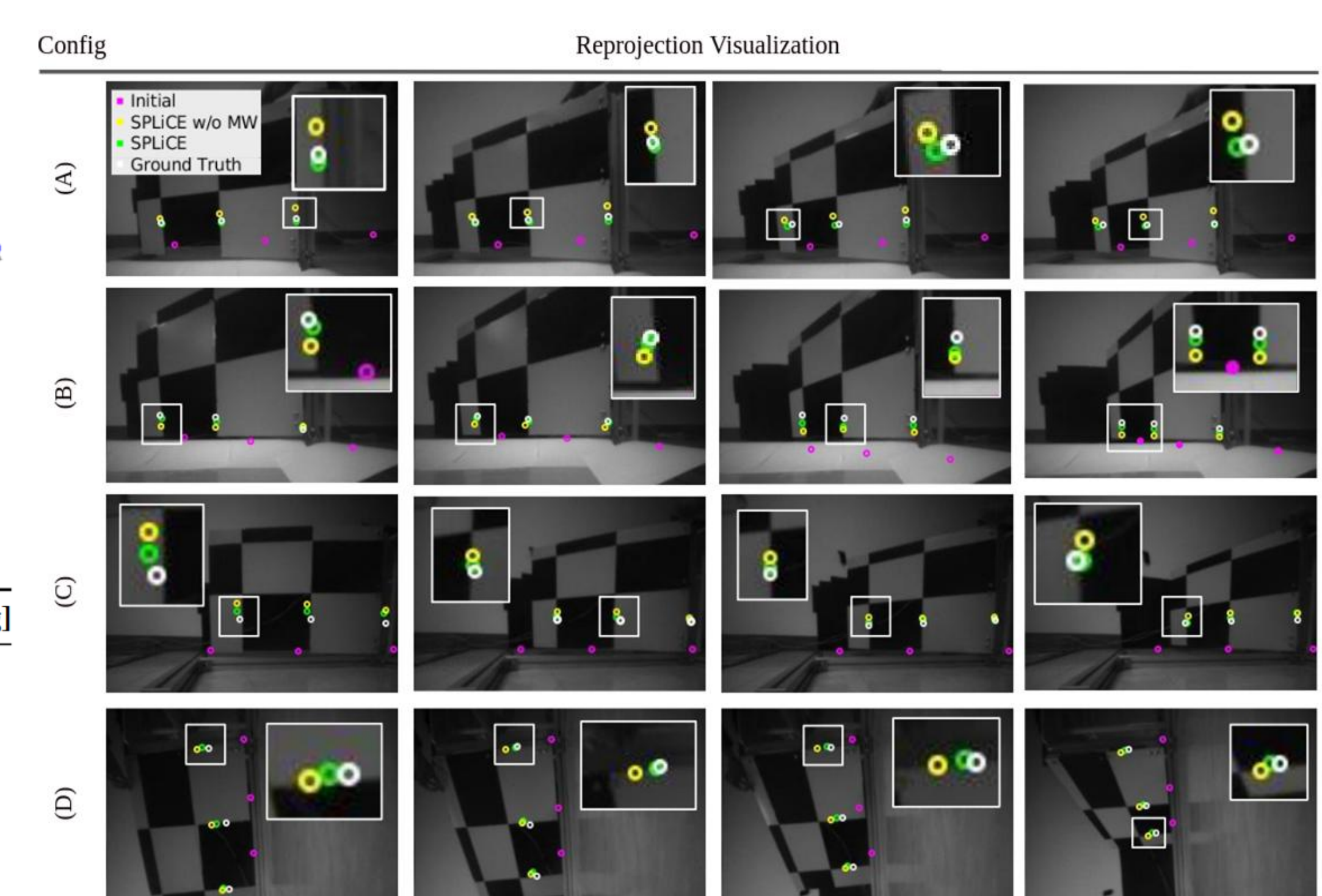
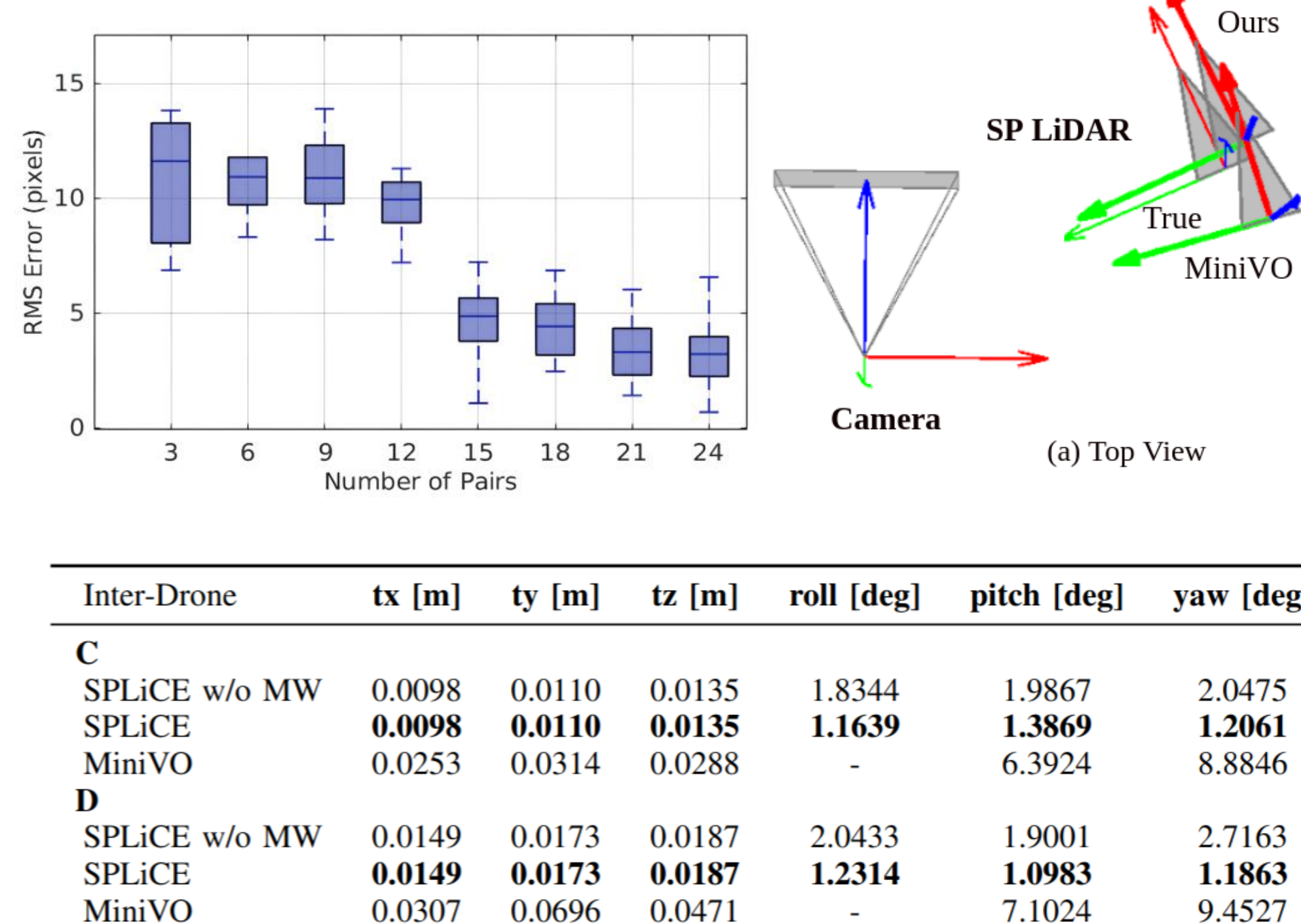
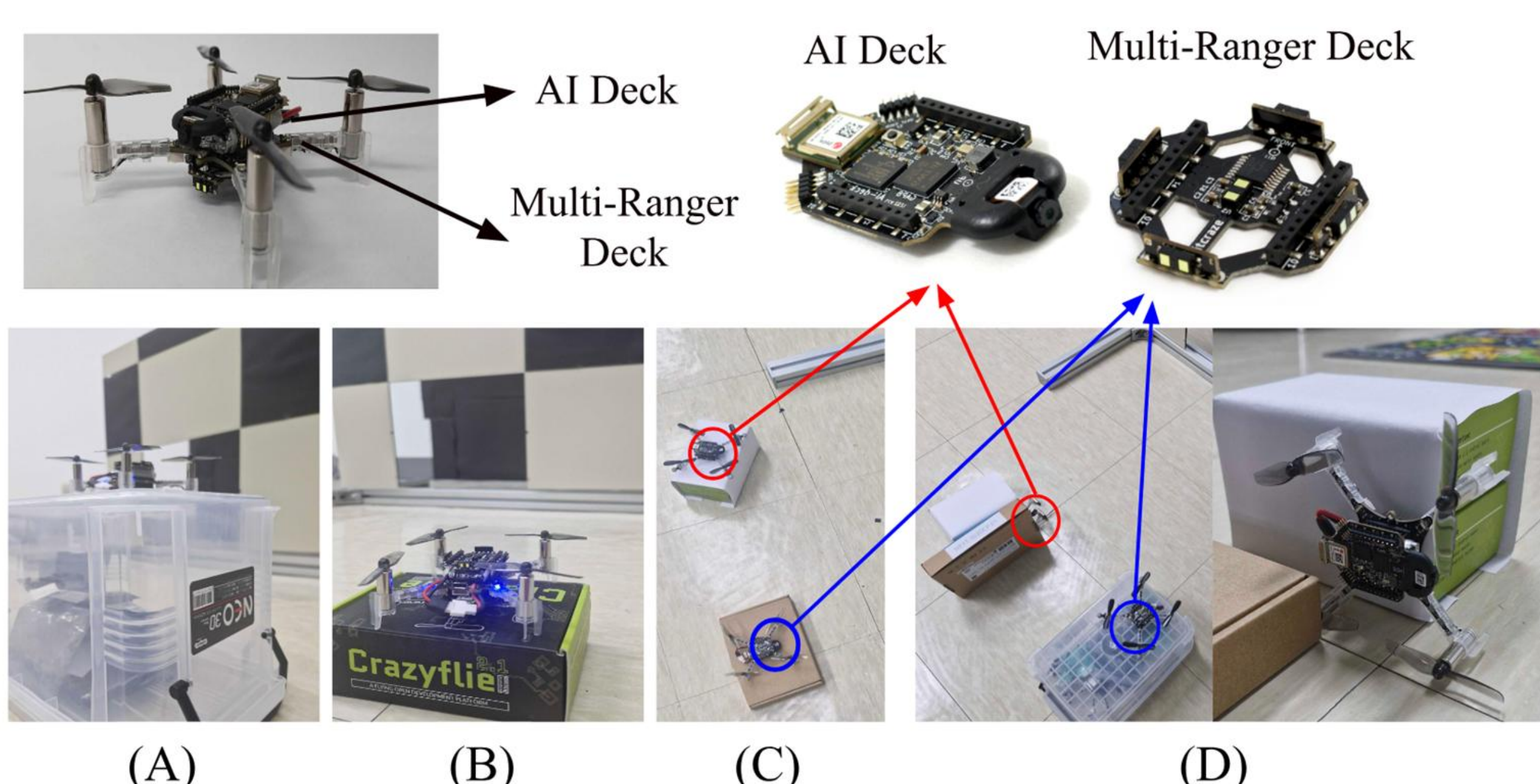
$$R^*, t^* = \arg \min_{R, t} \sum_{i=1}^N \left(\sum_{k=1}^3 (d_k^i)^2 + \sum_{j=1}^3 (p_c^i - p_j^i)^2 \right) \quad p_c^i = \left(\frac{1}{3} \sum_{j=1}^3 u_j^i, \frac{1}{3} \sum_{j=1}^3 v_j^i \right)$$

(N: number of image-LiDAR pairs)



Evaluation

- Four Experimental Setups: **intra- and inter-drone**



- The proposed SPLiCE achieves sub<3 pixel reprojection accuracy and sub<1° rotation error across all configurations
- These results confirm its effectiveness for reliable single-point LiDAR-camera calibration on lightweight micro-drone platforms